Fragility Paradox in Global Aviation: Structure vs Operational Risk

Presented by:

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Course:

Data Mining

Workflow overview

1 **Data Engineering**

Data collection and integration of operational data from diverse sources to create a complete dataset.

2 **Topological Validation**

Verification of the airport network structure and analysis of its connectivity and dependencies.

3 **Health Profiling**

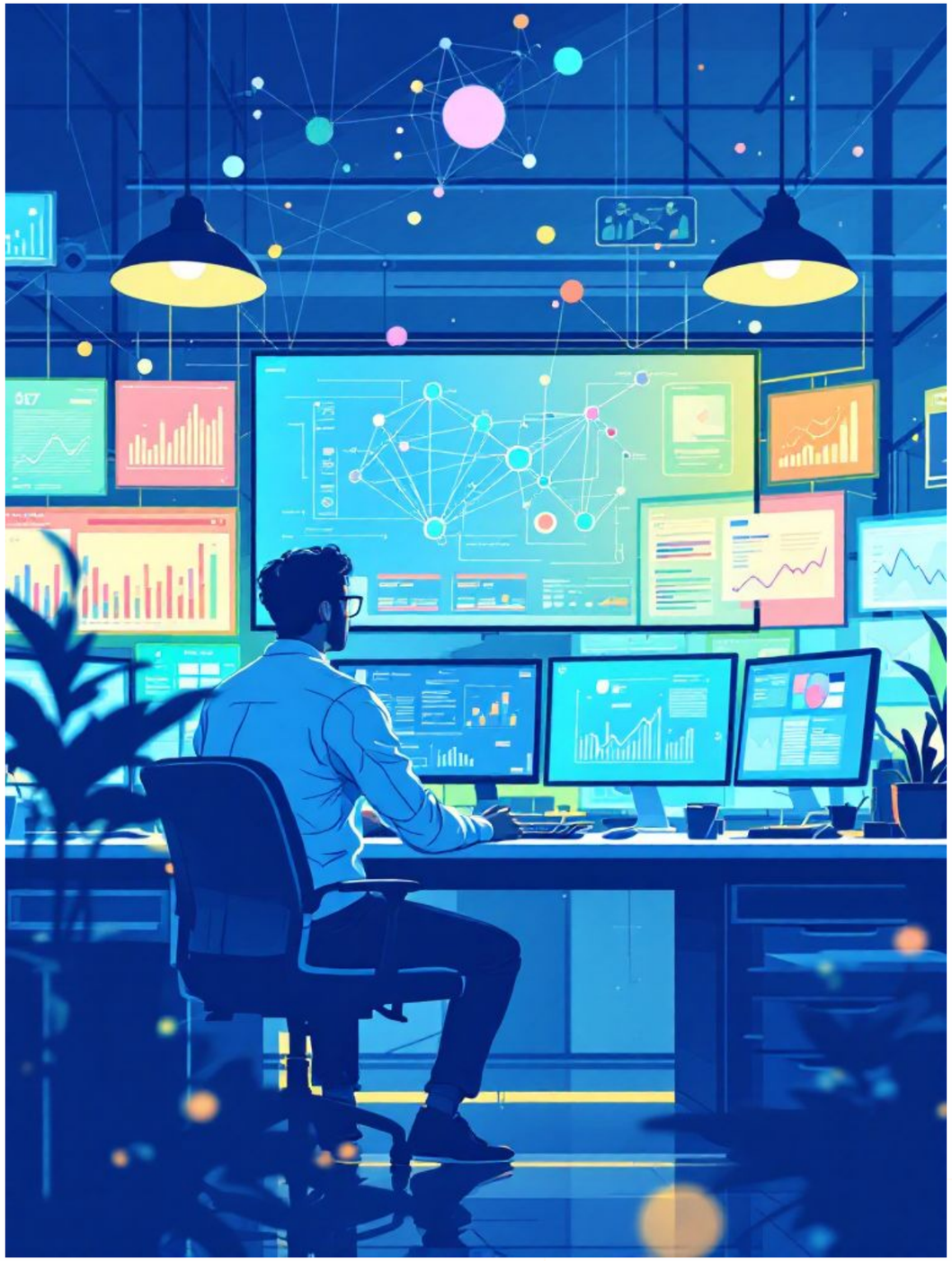
Unsupervised analysis to identify unique operational health profiles for each airport.

4 **Smart Switch**

Formulation of prescriptive recommendations to optimize risk management and improve resilience.

5 **Conclusions**

Synthesis of findings and discussion of the operational and strategic implications for the aviation sector.

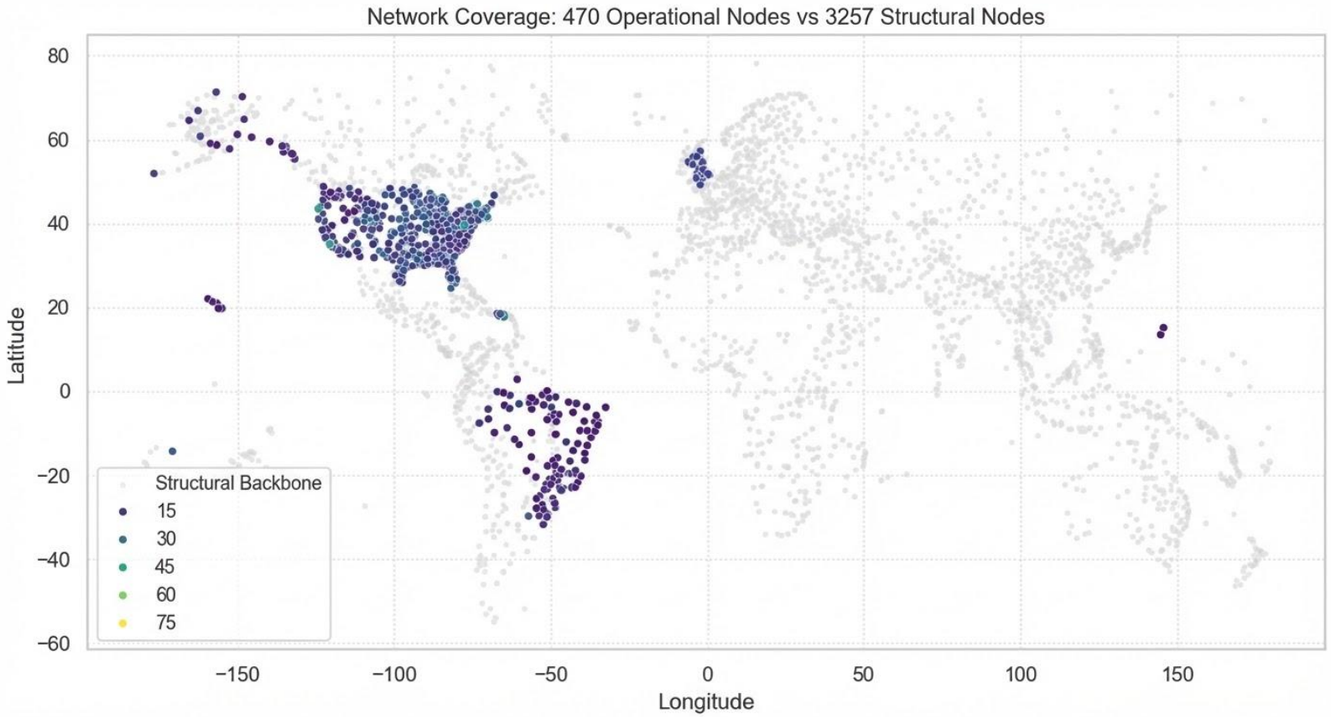


Data engineering – Building the dual graph

Structural Backbone Graph - **OpenFlights** Dataset:

- Raw data: **6,071** airports, **67,652** routes
- Data cleaning: removed **729** invalid routes, merged **29,881** parallel edges, pruned isolates

→ Final structural graph: **3,257** nodes, **37,041** routes.



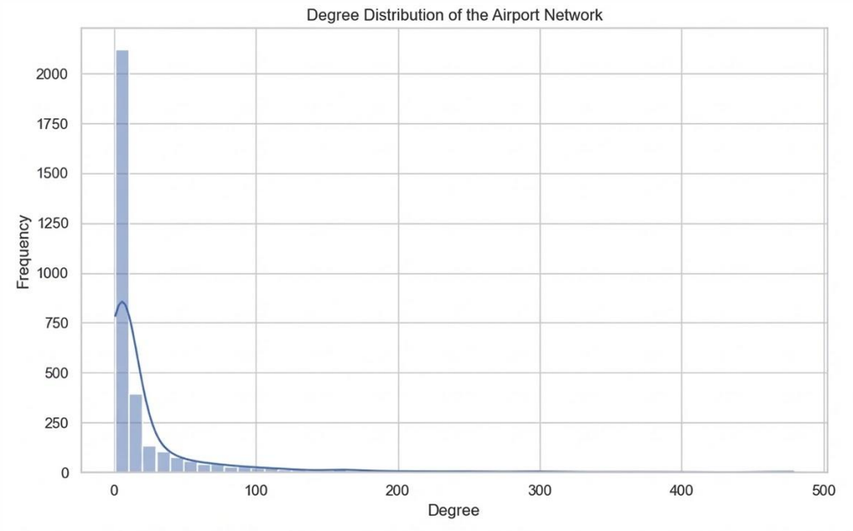
Operational layer - 2024 Delay Metrics from US **BTS**, Brazil **ANAC**, UK **CAA**:

- Origin-based filtering produced **470** reliably instrumented airports.
- Extracted **8,276** unique routes
- **38.9%** of these routes were missing in the structural network:
  - **3,072** routes (95.3% of missing) connect two existing airports
  - Added to structural graph

→ Final structural graph: **3,257** nodes, **40,113** routes.

→ Final operational layer: **470** nodes, **8,125** routes.

Topological validation – Scale-Free Property



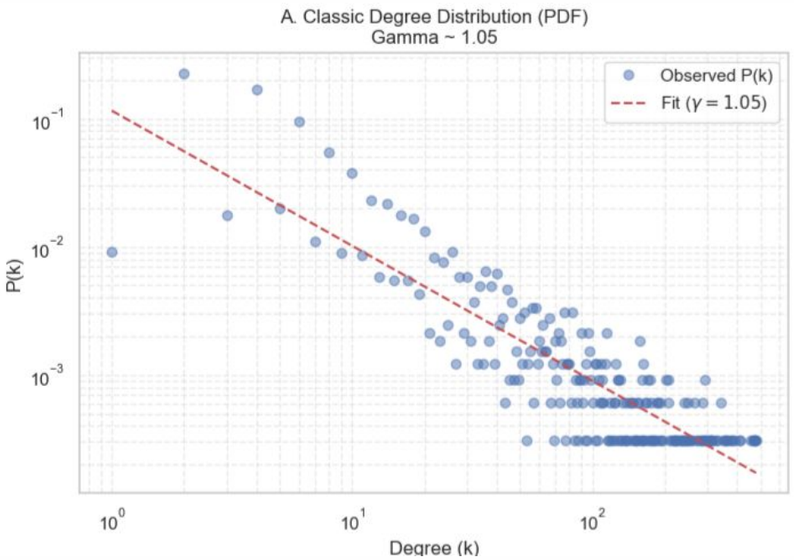
Degree distribution follows Power Law: $P(k) \sim k^{-\gamma}$

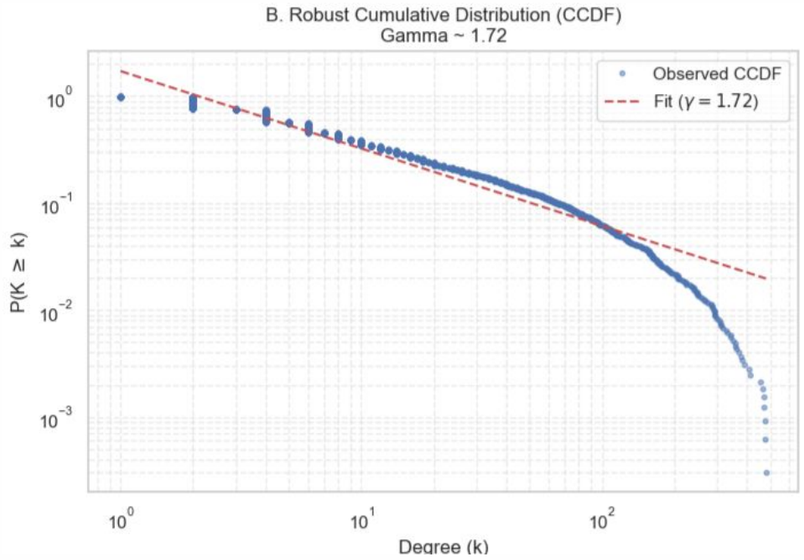
- PDF: $\gamma \approx 1.05$
- Robust CCDF: $\gamma \approx 1.72$
- Rank-Degree Plot: $\gamma \approx 1.78$

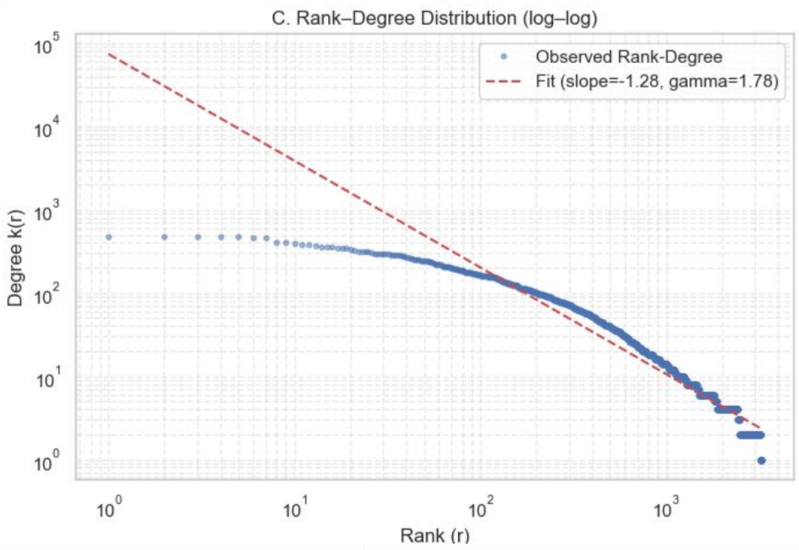
Directed Assortativity:

In-In
In-Out
Out-In
Out-Out

range ≈ -0.04



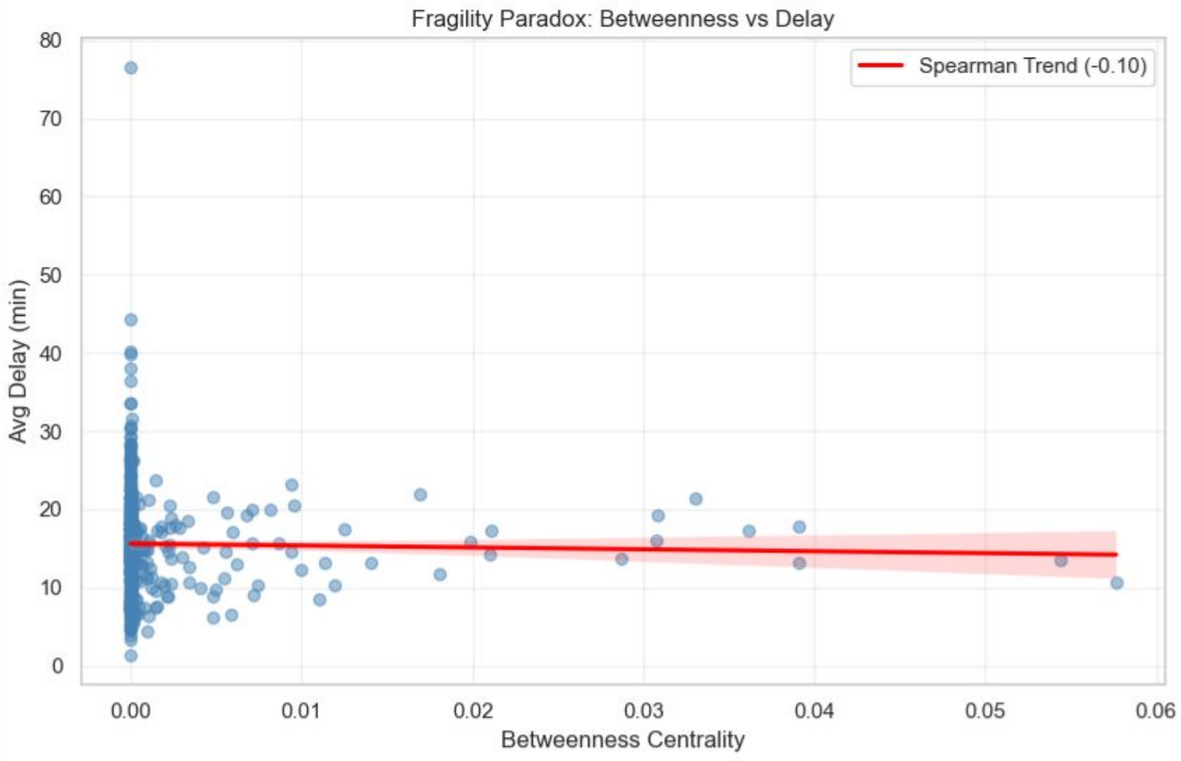




Robustness simulations

Testing the Fragility Paradox:

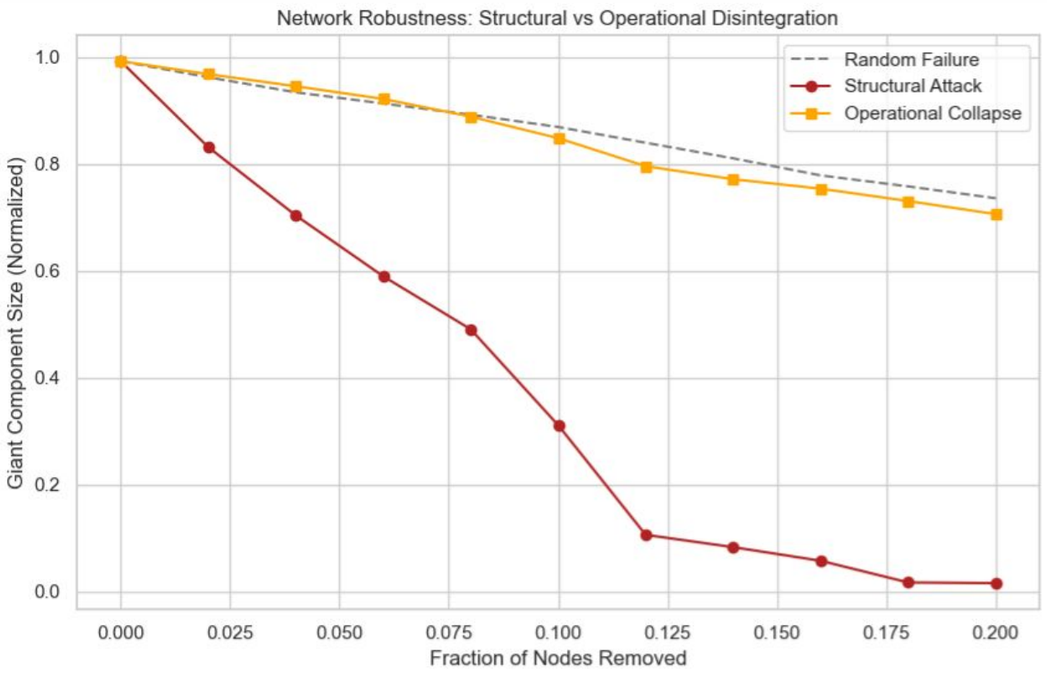
- Pearson Correlation: −0.0220
- Spearman Correlation: −0.1029



Three experiments:

1) Random failure
   → **gradual decline**
2) Targeted removal of high-centrality hubs
   → **rapid structural collapse**
3) Removal of highest-delay airports
   → connectivity loss ≈ random failure

Conclusion: operationally 'sick' nodes are not structurally central to the backbone.

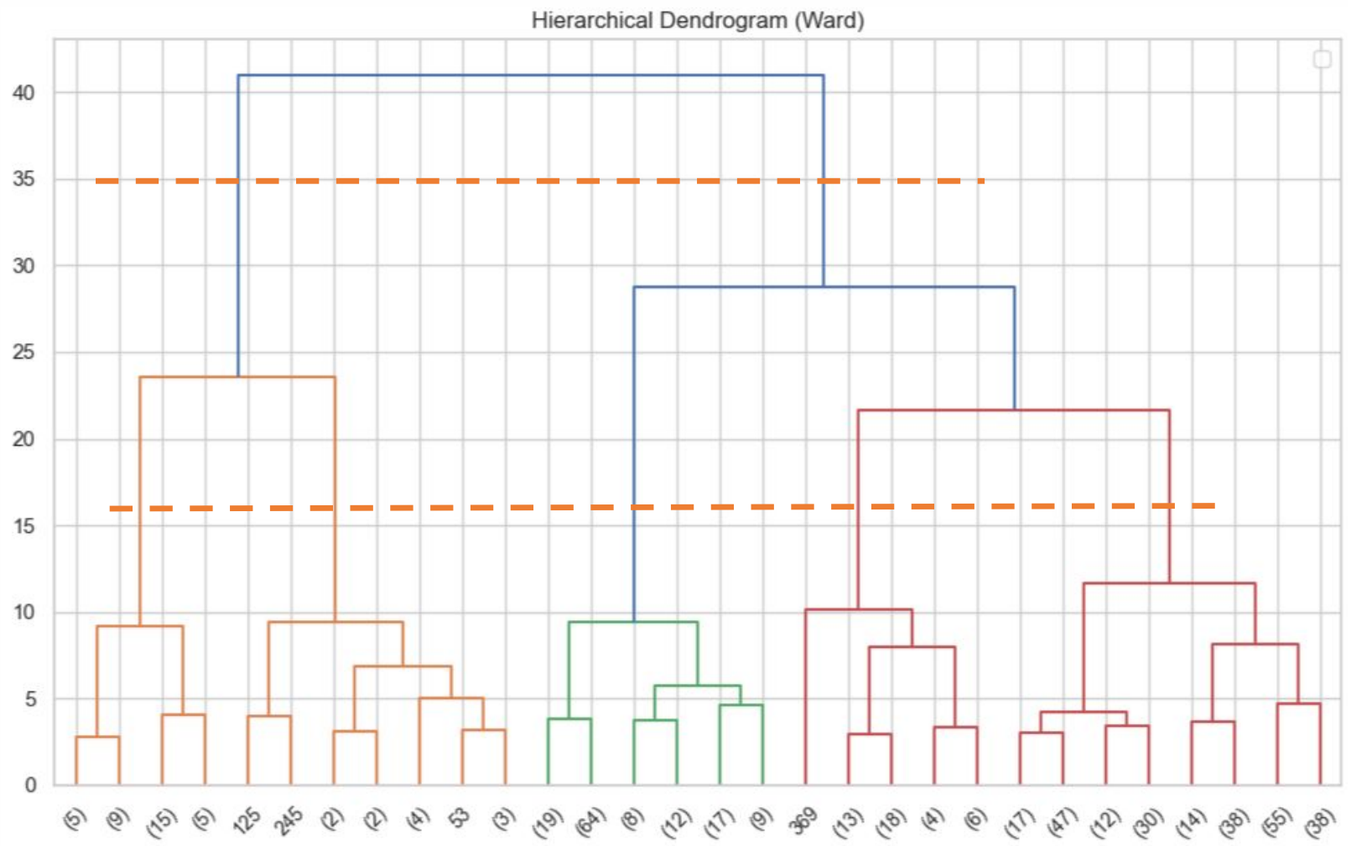


Airport health profiling (unsupervised learning)

5-dimensional feature space for the 470 airports:

- Features: PageRank, Degree, Betweenness Centrality
- Operational Features: Average Delay, Delay Variance

Important: values have different scales → Standardization



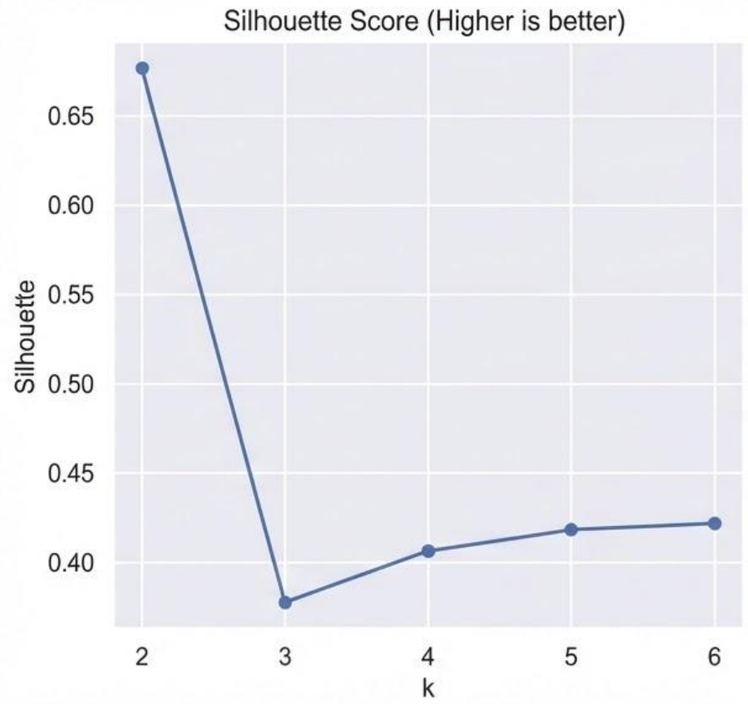
Airport health profiling (unsupervised learning)

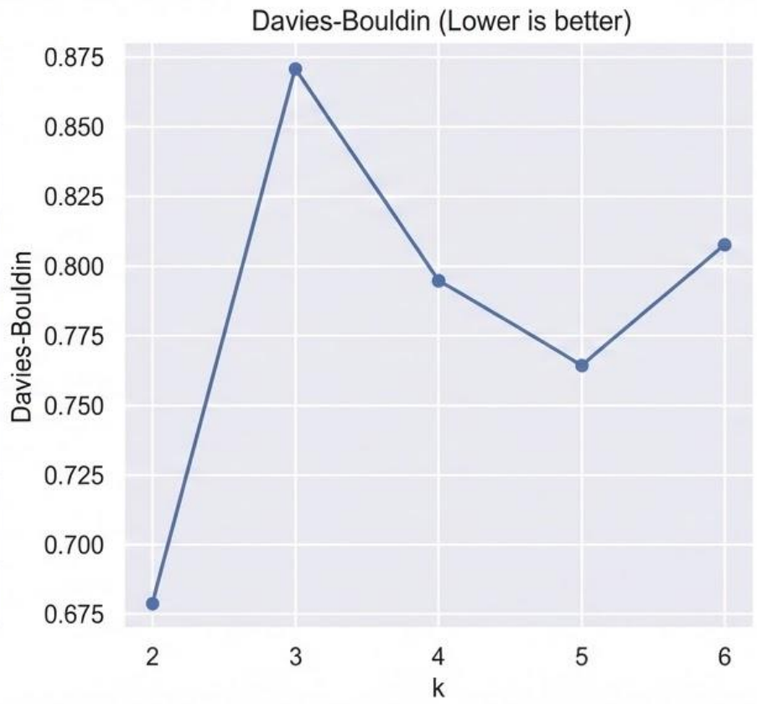
Test different metrics for **k ∈ [2, 6]**

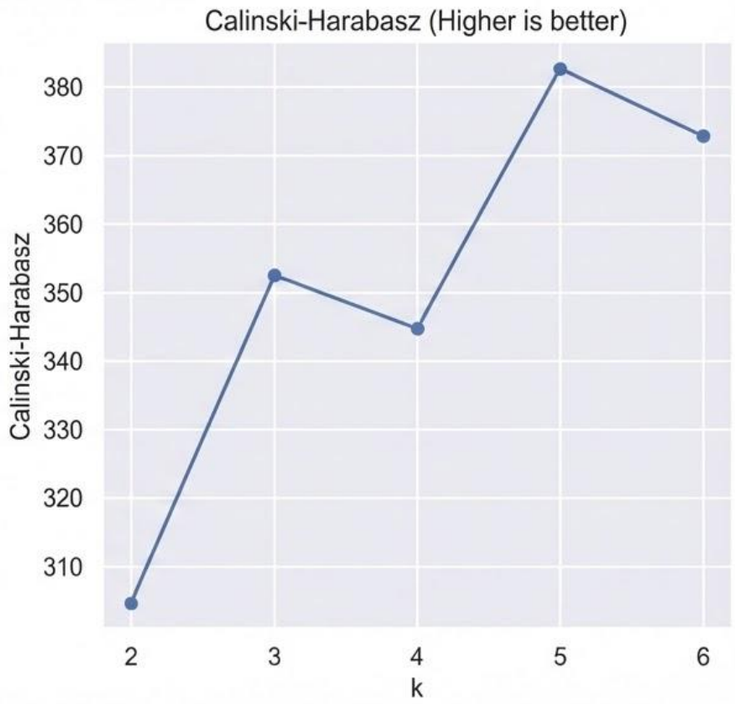
Silhouette Score: k = 2 → well separated clusters

Davies-Bouldin Index: k = 2 → still well separated clusters
k = 5 → also perform better than others

Calinski-Harabasz Index: k = 5 → really compacted clusters
k = 2 → less dense clusters







Clustering profiling

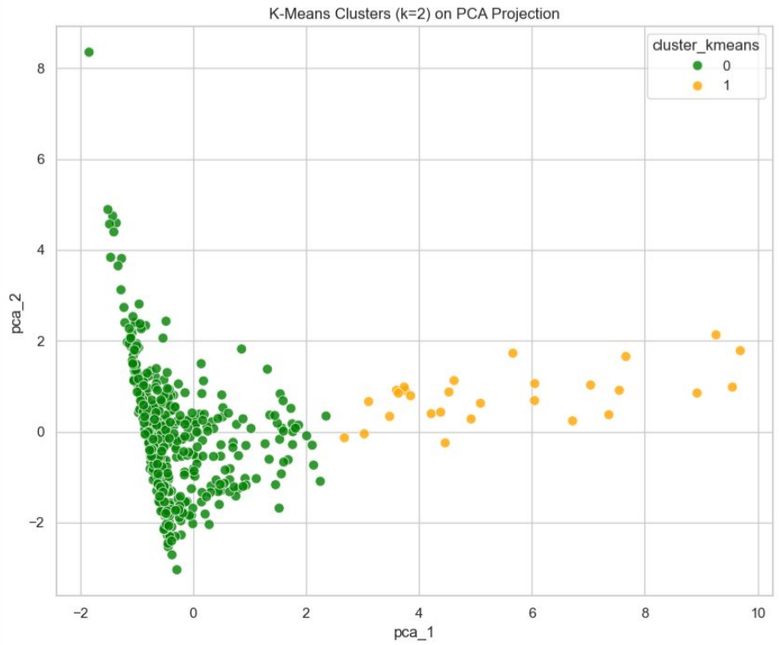
K-means clustering for: **k = 2**

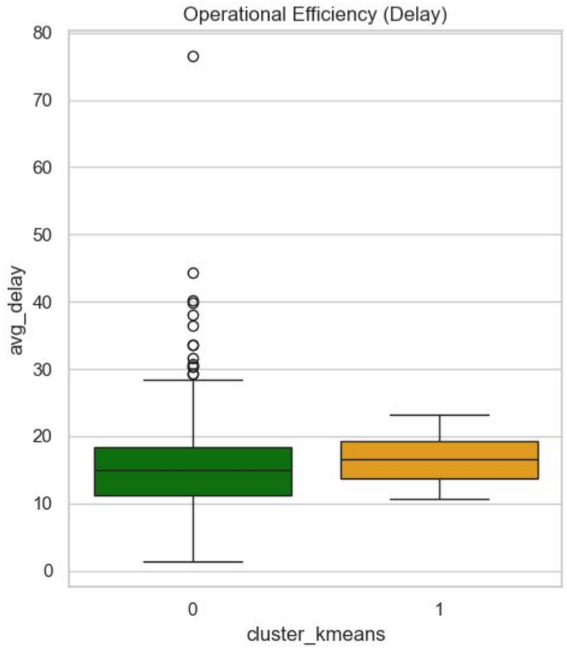
Cluster 0:

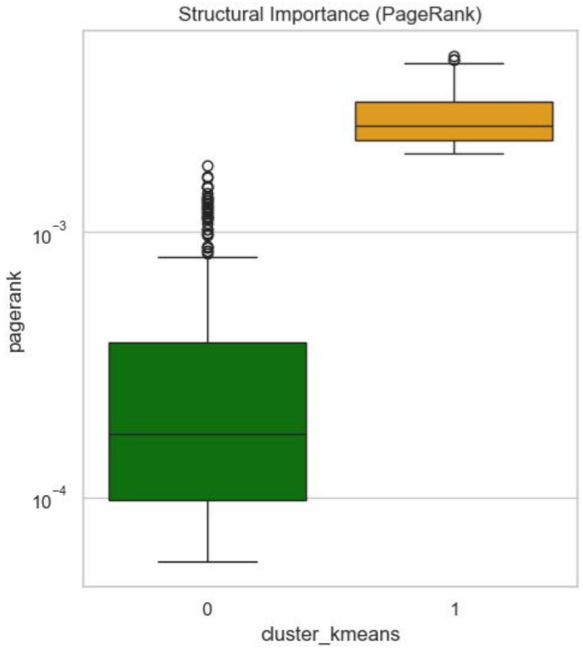
- Majority of airports
- Low Degree and PageRank
- Median delay is slightly lower, bigger variance

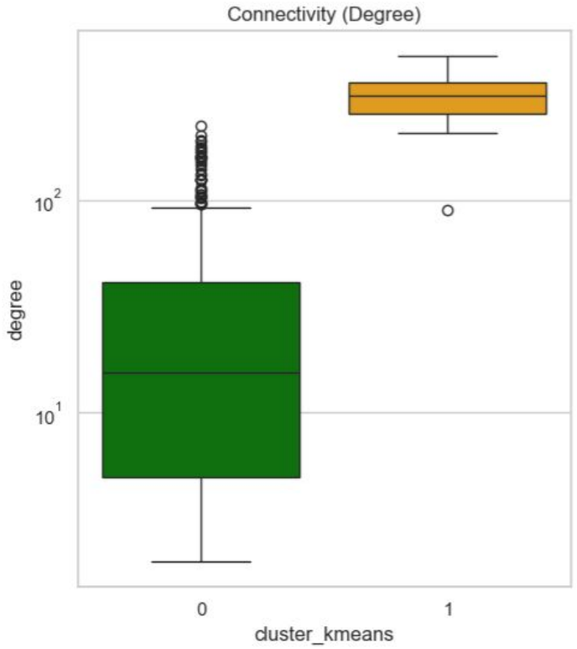
Cluster 1:

- High Degree and PageRank
- Slightly higher median on delays









Clustering profiling

K-means clustering for: **k = 5**

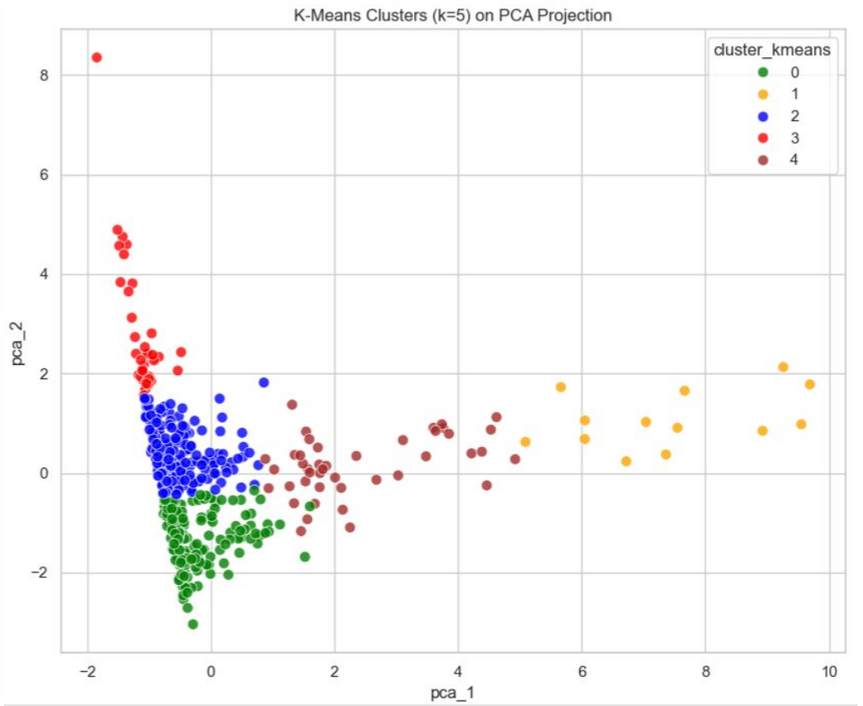
Cluster 0:
- Best performance
- Low Degree and PageRank

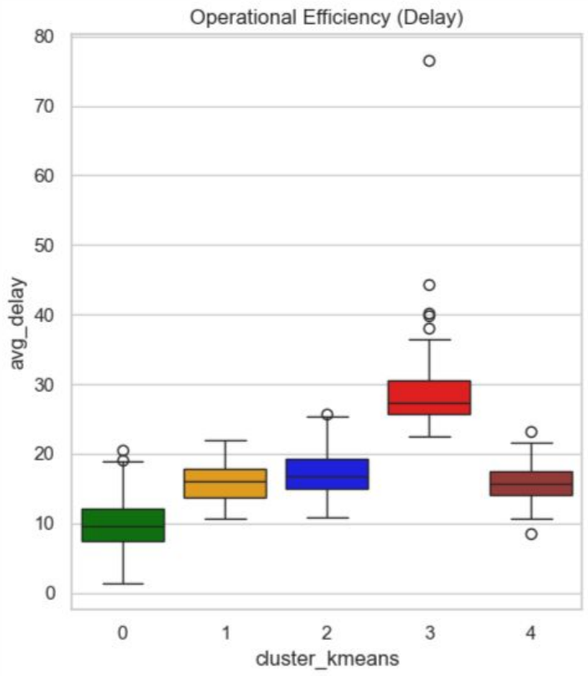
Cluster 1:
- High Degree and PageRank
- Low stable delays

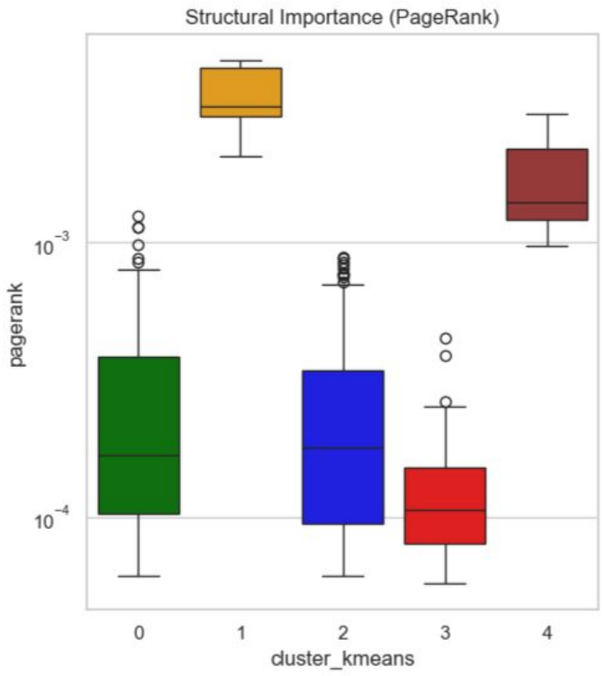
Cluster 2:
- Low Degree and PageRank
- Low stable delays

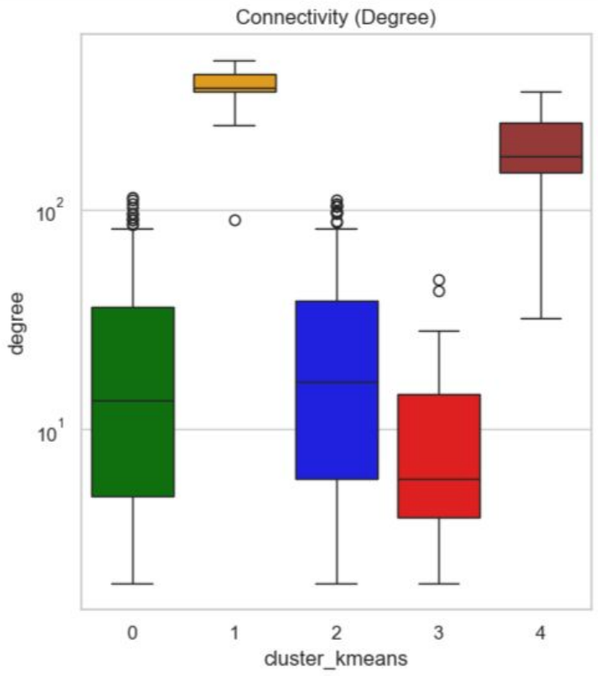
Cluster 3:
- Lowest Degree and PageRank
- Highest average delay

Cluster 4:
- High Degree and PageRank
- Low stable delays









Smart Switch: recommendation system

MinHash based on similarity of outgoing destinations

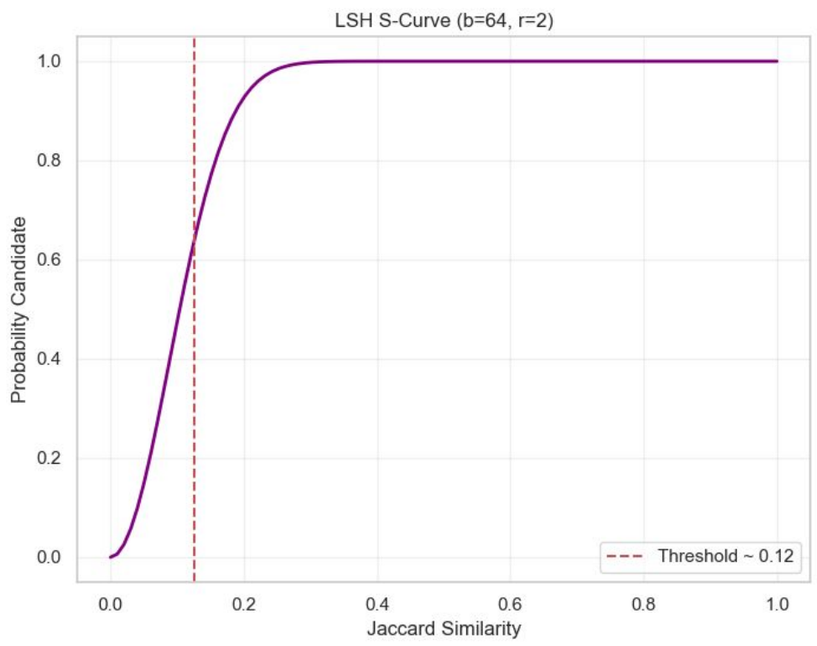
LSH configuration

128 permutations · 64 bands · threshold ≈ 0.12

→ Recall ≈ 92% · 17× speedup

Business filters

- Capacity ≥ 50% of degree
- Delay reduction ≥ 5 minutes
- Jaccard similarity > 0.4

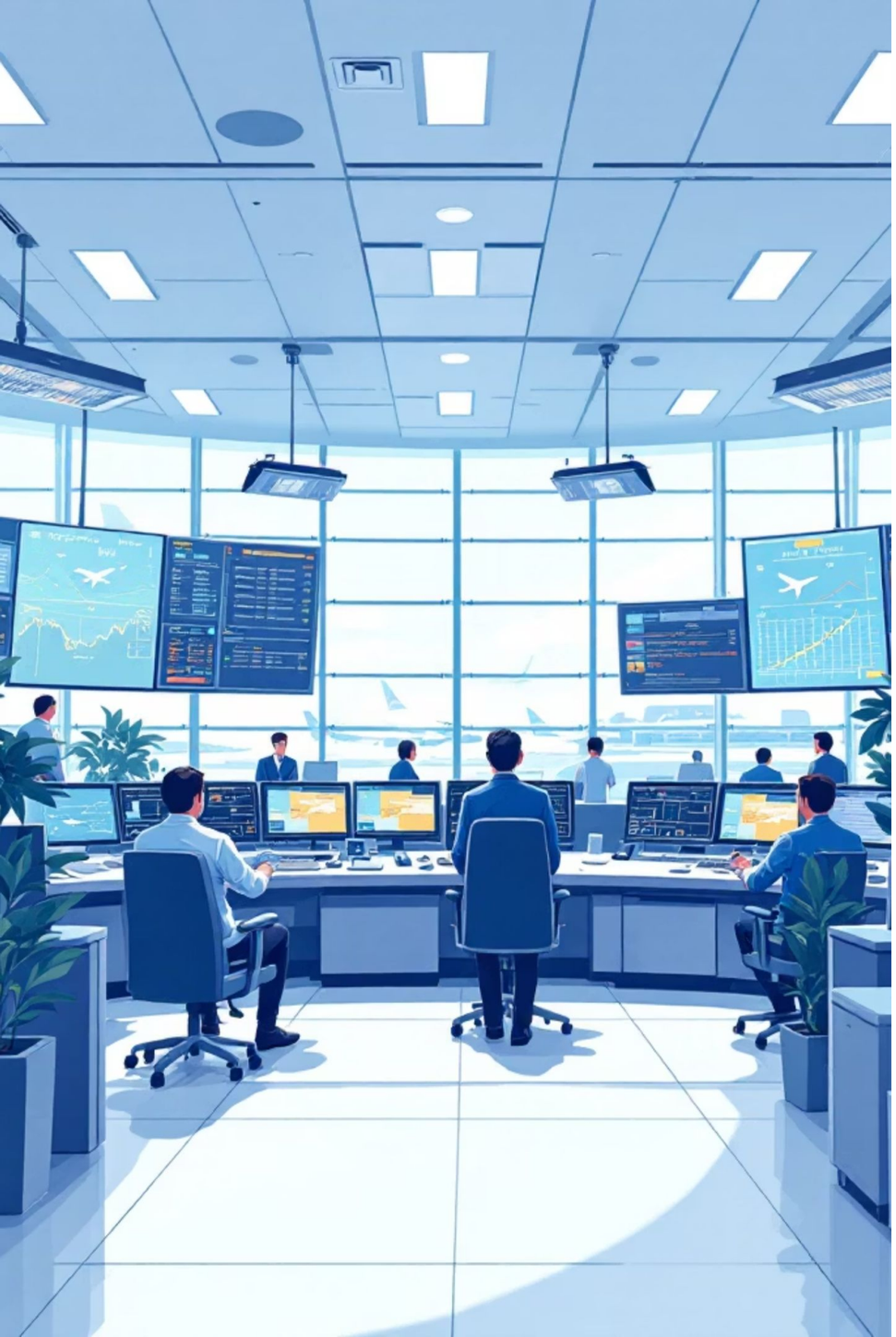


k = 2 (hub optimisation)

- 20 validated recommendations
- average gain ≈ 6.5 minutes
- Small gains per-airport
- Hub-to-Hub recommendations

k = 5 (peripheral bottlenecks)

- 157 validated recommendations
- average gain ≈ 13.8 minutes
- extreme cases > 60 minutes
- Small airports (avg capacity <10)



Conclusions & operational implications

- A central airport in the network is not automatically operationally fragile
- Hubs are robust; peripheral nodes drive volatility
- Prioritise targeted interventions at regional bottlenecks rather than focusing solely on large hubs.

Research next steps:

- Extend dataset across seasons and new countries

Thanks for the attention

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